UNIVERSITAS HARKAT NEGERI

Dosen Pengampu: Zaenul Arif, S.Kom., M.Kom · Mata Kuliah: Pemrograman Web Service (5537344)

22. Tugas Kelompok Minggu ke-4: API Design Document v1 (Milestone M2 - API Blueprint)

A. Identitas Proyek & Konvensi

Nama Project	BengkelMotor (Sistem Informasi Layanan Penjadwalan Servis dan Manajemen Riwayat)
Mata Kuliah / Kelompok	Pemrograman Web Service / 3 SKS · Kelompok 2 · Semester 7 (Ganjil) 2026/2027
Deskripsi	Platform layanan bengkel motor terintegrasi dengan penjadwalan servis instan (auto-confirm), kontrol konkurensi (pessimistic locking anti-overbooking), logbook digital mekanik, dan pelacakan riwayat perawatan.
Versi API & Konvensi	v1 (prefix /api/v1) · Inggris (resource/atribut snake_case), pesan respon bahasa Indonesia

Anggota Tim & Tanggung Jawab Modul:

No	Nama Mahasiswa	NIM	Tanggung Jawab Modul / Fitur
1	Intan Komalasari	23215012	Auth & User Profile Module (Sanctum JWT, RBAC 3 Role)
2	Feldi Sanjaya	23215022	Vehicle Fleet Module (Manajemen Motor, Ownership Policy)
3	Imzy Zulijar Setiawan	23215035	Service Slot & Schedule Module (Kapasitas & Kuota Real-Time)
4	Eko Saputro	23215050	Booking & Concurrency Module (Pessimistic Locking Auto-Confirm)
5	Zakky Fawwaz Mubarok	23215061	Service Record & Logbook Module (Tindakan, Odometer, Suku Cadang JSON)

B. Pemodelan Resource & Relasi (Total 10 Resource)

No	Resource	Modul	Atribut Utama & Tipe Data	Jenis
1	**users**	Auth	id (bigint), nama (string), email (string, unik), no_hp (string), role (enum), created_at	Collection
2	**me**	Auth	Data profil dan token pengguna yang sedang terautentikasi	Singleton
3	**vehicles**	Vehicle	id (bigint), user_id (bigint), no_polisi (string, unik), merk, model, tahun	Collection
4	**service_slots**	Slot	id (bigint), tanggal (date), jam_mulai (time), kuota_maksimal, terisi, is_available	Collection
5	**bookings**	Booking	id (bigint), booking_code (unik), user_id, vehicle_id, slot_id, keluhan, status	Collection
6	**service_records**	Record	id (bigint), booking_id (unik), mechanic_id, odometer_km, tindakan, suku_cadang	Collection
7	**parts**	Record	nama (string), qty (integer), keterangan (string)	Sub-resource
8	**vehicle_history**	Record	Agregasi riwayat servis kronologis berdasarkan no_polisi motor	Sub-resource
9	**mechanic_tasks**	Booking	Antrean motor siap servis di pit bengkel berstatus CONFIRMED	Sub-resource
10	**cancellations**	Booking	Data audit pembatalan booking beserta alasan dan pemulihan kuota	Sub-resource

C. Endpoint Matrix (25 Endpoint Representatif)

No	Method	Endpoint	Fungsi	Aktor	Sukses	Gagal
1	POST	/api/v1/auth/register	Mendaftarkan akun baru	Public	201	409, 422
2	POST	/api/v1/auth/login	Login & terbitkan token	Public	200	401, 422
3	POST	/api/v1/auth/logout	Mencabut token aktif	Semua (Login)	200	401
4	GET	/api/v1/me	Melihat profil sendiri	Semua (Login)	200	401
5	PATCH	/api/v1/me	Mengubah profil sendiri	Semua (Login)	200	401, 422
6	GET	/api/v1/users	Melihat master user	Admin	200	401, 403
7	GET	/api/v1/vehicles	Melihat motor sendiri	Customer	200	401
8	POST	/api/v1/vehicles	Mendaftarkan motor baru	Customer	201	401, 409, 422
9	GET	/api/v1/vehicles/{id}	Melihat detail motor	Customer, Admin	200	401, 403, 404
10	PUT	/api/v1/vehicles/{id}	Mengganti data motor	Customer (Owner)	200	401, 403, 404, 422
11	DELETE	/api/v1/vehicles/{id}	Menghapus data motor	Customer (Owner)	200	401, 403, 404, 409
12	GET	/api/v1/service-slots	Katalog slot real-time	Public / Cust	200	422
13	POST	/api/v1/service-slots	Membuat slot jam baru	Admin	201	401, 403, 409, 422
14	PUT	/api/v1/service-slots/{id}	Perbarui kuota slot	Admin	200	401, 403, 404, 422
15	DELETE	/api/v1/service-slots/{id}	Hapus slot belum terisi	Admin	200	401, 403, 404, 409
16	POST	/api/v1/bookings	Booking Auto-Confirm & Lock	Customer	201	401, 403, 404, 409, 422
17	GET	/api/v1/bookings	Riwayat booking servis	Customer, Mech, Adm	200	401
18	GET	/api/v1/bookings/{id}	Detail transaksi booking	Customer, Mech, Adm	200	401, 403, 404
19	PATCH	/api/v1/bookings/{id}/cancel	Batal booking & pulihkan kuota	Customer, Admin	200	401, 403, 404, 409
20	GET	/api/v1/mechanic/bookings	Antrean motor di pit	Mechanic	200	401, 403
21	POST	/api/v1/service-records	Rekam logbook servis	Mechanic	201	401, 403, 404, 409, 422
22	GET	/api/v1/service-records/{id}	Detail logbook spesifik	Customer, Mech, Adm	200	401, 403, 404
23	GET	/api/v1/bookings/{id}/records	Riwayat pengerjaan booking	Customer, Mech, Adm	200	401, 403, 404
24	GET	/api/v1/service-history	Lacak servis via plat nomor	Public / Cust	200	422
25	GET	/api/v1/health	Status server & DB	Public	200	500

D. Desain Query: Filtering, Searching, Sorting & Pagination

• GET /api/v1/service-slots?tanggal=2026-10-06&tersedia;=true&sort;=jam_mulai&page;=1&per;_page=15 (Maks. 50)
• GET /api/v1/bookings?status=CONFIRMED&q;=BK-2026&sort;=-created_at&page;=1&per;_page=15 (Scope per role)

E. Desain Representasi Data (Item, Collection, Create)

```
// 1. Item: GET /api/v1/bookings/1 -> 200 OK
{
  "status": "success", "message": "Detail reservasi servis berhasil diambil",
  "data": {
    "id": 1, "booking_code": "BK-20261004-7A9B", "status": "CONFIRMED",
    "vehicle": { "id": 1, "no_polisi": "G 2222 KL", "merk": "Yamaha", "model": "Aerox 155" },
    "slot": { "id": 1, "tanggal": "2026-10-06", "jam_mulai": "09:00:00", "kuota_maksimal": 5, "terisi": 1 }
  },
  "meta": { "timestamp": "2026-10-04T15:10:00+07:00", "version": "v1" }
}

// 2. Collection: GET /api/v1/service-slots?page=1&per;_page=2 -> 200 OK
{
  "status": "success", "data": [
    { "id": 1, "jam_mulai": "08:00:00", "kuota_maksimal": 5, "terisi": 1, "is_available": true },
    { "id": 2, "jam_mulai": "09:00:00", "kuota_maksimal": 5, "terisi": 5, "is_available": false }
  ],
  "links": { "first": "...?page=1", "last": "...?page=4", "prev": null, "next": "...?page=2" },
  "meta": { "current_page": 1, "per_page": 2, "total": 8, "version": "v1" }
}
```

F. Error & Status Code Standar

```
// 422 Validasi Gagal: POST /api/v1/bookings -> 422 Unprocessable Content
{
  "status": "error", "message": "Data yang dikirim tidak valid.", "code": "VALIDATION_FAILED",
  "errors": { "vehicle_id": ["Bidang vehicle_id wajib diisi."], "slot_id": ["Slot tidak valid."] },
  "meta": { "timestamp": "2026-10-04T15:25:00+07:00", "version": "v1" }
}

// 409 Aturan Bisnis: POST /api/v1/bookings -> 409 Conflict
{
  "status": "error", "message": "Slot servis pada jam yang dipilih sudah penuh.", "code": "SLOT_CAPACITY_EXCEEDED",
  "meta": { "timestamp": "2026-10-04T15:25:00+07:00", "version": "v1" }
}
```

G. Aksi Non-CRUD & Catatan Keputusan Desain

• **Auto-Confirm & Pessimistic Locking**: POST /api/v1/bookings mengunci baris slot (lockForUpdate()) memeriksa kuota aktual secara atomik tanpa race condition.
• **Pembatalan Booking**: PATCH /api/v1/bookings/{id}/cancel memulihkan kuota slot secara otomatis dan mencatat audit trail status CANCELLED.
• **Suku Cadang JSON**: Disimpan pada service_records.suku_cadang demi efisiensi fleksibilitas entri komponen sparepart.

Lampiran: Hasil Peer Review

No	Butir Pemeriksaan	Ya	Tidak	Catatan Reviewer	Perbaikan
1	URI kata benda jamak, huruf kecil, tanpa kata kerja	[v]	[ ]	Sesuai konvensi RESTful	Tidak ada
2	Prefix /api/v1 dipakai konsisten	[v]	[ ]	Sangat konsisten	Tidak ada
3	Method sesuai operasi; GET tidak mengubah data	[v]	[ ]	Idempotent & Safe terpenuhi	Tidak ada
4	POST -> 201, DELETE -> 200/204, status gagal ada	[v]	[ ]	Header Location disertakan	Tidak ada
5	Nesting maksimal 2 tingkat	[v]	[ ]	Terjaga (/bookings/{id}/records)	Tidak ada
6	Collection memiliki pagination & batas maksimum	[v]	[ ]	Page-based + meta links	Tidak ada
7	Filter/sort memakai query parameter whitelist	[v]	[ ]	Aman dari query injection	Tidak ada
8	Format error seragam	[v]	[ ]	Struktur error terstandarisasi	Tidak ada